

System Methodology & Technical Architecture

Deterministic RAG Scoring Engine, Natural Language Synthesis & System Architecture

1. RAG Determination Methodology

The core philosophy of the system is deterministic, auditable scoring. Instead of delegating RAG classification to black-box LLMs - which are susceptible to hallucination, inconsistency, and lack mathematical validation - the RAG classification is computed via a structured rule engine. The LLM acts as an explanatory and narrative summarization layer on top of the verified metrics.

1.1 Weighted Signals Model

Signal Layer	Weight	Computation & Normalization Logic
Schedule Slippage	30%	Scans schedule health column for explicit 'Red'/'Amber' flags; checks total float erosion and counts late active tasks.
Progress Gap	20%	Compares actual percent complete against the expected progress linearly calculated from elapsed timeline.
Milestone Health	20%	Analyzes incomplete key milestones, measuring the ratio of delayed/high-risk milestones to completed ones.
Blockers & Float	15%	Identifies critical path tasks, negative total float, and checks task comments/predecessors for blocker keywords.
Sentiment Theme	10%	Evaluates the emotional tone, outlook, and thematic urgency of PM comments on a scale from 0 to 100.
Budget Burn	5%	Calculates cost variance and burn rates. Excluded if budget data is missing (weights re-normalize to remaining 95%).

1.2 Scoring Normalization & Mapping

Overall project health is a weighted sum divided by total active weights. If budget data is absent, its weight is set to 0, and the remaining weights are normalized to sum to 1.0. The project score is mapped to the final RAG status as follows:

- Green (Healthy):** Score 0 to 34 and no critical override triggered.
- Amber (Watch-list):** Score 35 to 69.
- Red (High-risk):** Score 70 to 100, or if a critical override is active.

1.3 Critical Override Rules

To capture severe systemic risks that a weighted average might dilute, the engine applies overrides to force Red:

- 1.** Severe Critical Path Delay: An incomplete critical path task delayed by > 15 working days.
- 2.** Milestone Failure: Key milestones missing baseline finish dates by > 30 days and remaining incomplete.
- 3.** Multiple Blockers: More than 3 active tasks flag-marked as blocked by external dependencies.
- 4.** Widespread Red Schedule: Over 40% of active tasks flagged as 'Red' or 'Yellow' schedule health.

2. Technical Stack

Backend API & Processing Layer:

- Python 3.10+: Static type annotations and strict coding patterns.
- FastAPI & Uvicorn: High-performance asynchronous REST API backend (configured on port 8001).
- SQLite Database: Re-writes, stores and queries snapshots, normalized tasks, comments, and RAG logs.
- openpyxl: Robust Excel reader with safety checks and explicit file closures preventing Windows file locks.
- python-pptx: Slide generator compiling widescreen monthly portfolio decks.
- LLM API (Groq/Gemini): Processes structured JSON schemas for narratives, sentiment summaries, and themes.

Frontend Dashboard Client:

- React 19 & TypeScript: Modern single-page application framework.
- Vite: Instant compilation and hot module reloading (dev server on port 3000).
- TanStack Router: Declarative type-safe routing.
- Tailwind CSS & Shadcn UI: Polished slate-themed responsive interface layouts.
- TanStack Query: Optimized server-state caching and asynchronous query orchestration.

3. Product Features

- Resilient Ingestion: Hierarchy mapping using 'Level' or 'Ancestors' indentation trees.
- Project Stage Fallback: Scans task nodes to find the current active phase if omitted in the summary.
- Explainability Core: Renders details under 'Why this status' and 'Signals' citing row numbers.
- Full Preservation: Renders every original workbook column raw attribute under 'Preserved attributes'.
- Switcher & Deletion: Header project selector and cascade deletion of snapshot records and output files.

4. Data Lifecycle

1. UPLOAD: React dropzone sends .xlsx workbook to backend.
2. NORMALIZE: Python maps column schemas, filters empty rows, and detects stages.
3. SCORE: Deterministic formulas evaluate metrics, and overrides are checked.
4. AI SUMMARY: Groq/Llama-3.3 summarizes narratives and sentiments.
5. PERSIST: Records are written to SQLite, and JSON/Markdown logs are stored.
6. PRESENT: UI loads data in tables/tabs, and allows PowerPoint generation.